Preparation

You will receive a pen with which you are going to trace the edge of a ruler taped to a trackpad.

Your hand and the trackpad are going to be covered during the task, so you won’t be able to see your hand.

Instead, at the beginning of each trial you will receive a vibrational signal via the vibrator taped to your hand. You will then use the intensity of the vibration to estimate the distance to the target/ obstacle.

Overview:

Our experiment consists of three phases:

1. Pre-Test

In this phase, you will have the opportunity to get familiar with the experimental conditions. However, you will not receive any feedback about your performance.

2. Training

In this phase, your task will be to either reach or avoid a target. You will receive accuracy feedback based on your performance. You will also have the chance to take short breaks during this phase before starting each block (there are 5 blocks in total). Feel free to relax a little!:)

3. Post-Test + Training

This final phase is designed to assess the learning effect. Just like the Pre-Test, you will not receive any feedback about your performance during this phase. However, there is going to be a short Training block in the middle of this phase, where you will be receiving accuracy feedback.

Instructions

Reaching task

Each trial will consist of one reaching or avoiding task

Your task will always be to interpret the vibrational signal you receive at the beginning of each trial and then estimate where the target is

1. Place the pen at the marked starting point.
2. At the beginning of each trial you will feel a vibration with its intensity corresponding to your distance to the target
3. Your task is to interpret the vibration you feel and then to determine your distance to the target.
4. Move the pen along the edge of one of the rulers.
5. Stop at the point you believe to have hit the target.

Pre-Test

Reaching Group

You will now have a few trials to get familiar with your task.

Your first task will be to find the target point based on your interpretation of the vibration.

1. Place the pen at the staring point and choose one of the rulers to follow.
2. You will then feel the vibration.
3. After you have felt the vibration you need to decide how far from the target you are
4. Move the pen according to your estimation and pause there to finish the trial

Pre-Test

Reaching Group

You are now already familiar with the reaching task.

Your next task will be to get familiar with the avoiding task.

The procedure is similar to your last task, but instead of hitting the target you now want to avoid an obstacle.

The intensity of the vibration will give you information on how far the obstacle is.

After you have felt the vibration you will again follow the edge of the ruler, but this time you are switching to the other edge, once you think you are close to reaching the obstacle.

To finish the movement you have to trace the second edge to the finish point (target).

Training:

Reaching Group

Now you have reached the training phase.

In this phase you will learn to interpret the vibrations correctly.

The training consists of 5 blocks. Between each block you can take a short break if you need.

After each trial you will now receive feedback on how well you performed.

You will now start training on the reaching task.

Post-Training:

Reaching Group

To see how well your training went you will now do some test trials.

We will start again with the reaching task. This time without any feedback.

Post-Training:

Reaching Group

To refresh your memory, we will have another block of training.

Again, doing the reaching task.

Post-Training:

Reaching Group

You are almost done!

To finish the experiment you are going to do the avoiding task again. Remember?

After you feel the vibration you need to follow the edge and switch sides to avoid the obstacle.

Pre-Test

Avoiding Group

You will now have a few trials to get familiar with your task.

Your first task will be to avoid an obstacle at the right time, based on your interpretation of the vibration.

1. Place the pen at the starting point and choose one of the rulers to follow.
2. You will then feel the vibration.
3. After you have felt the vibration you need to decide how far from the obstacle you are
4. Move the pen according to your estimation along the edge until the point where you think the obstacle is placed.
5. When you think you are close to the obstacle switch to the other edge and finish the task by moving your pen to the finish point (target).

Pre-Test

Avoiding Group

You are now already familiar with the avoiding task.

Your next task will be to get familiar with the reaching task.

The procedure is similar to your last task, but instead of avoiding an obstacle you now only want to find the target point.

The intensity of the vibration will give you information on how far away the target is.

After you have felt the vibration you will again follow the edge of the ruler, but this time you don’t need to switch sides.

To finish the task pause your movement when you think you have hit the target.

Training

Avoiding Group

Now you have reached the training part.

In this part you will learn to interpret the vibrations correctly.

The training consists of 5 blocks. Between each block you can take a short break if you need.

After each trial you will now receive feedback on how well you performed.

You will now start training on the avoiding-task.

Post-Training

Avoiding Group

To see how well your training went you will now do some test trials.

We will start again with the avoiding task. This time without any feedback.

Post-Training

Avoiding Group

To refresh your memory we will have another block of training.

Again doing the avoiding task.

Post-Training

Avoiding Group

You are almost done!

To finish the experiment you are going to do the avoiding task again. Remember?

After you feel the vibration you need to follow the edge and switch sides to avoid the obstacle.

Need a break?

Need a break?

Need a Break??

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